

TAUFIQUL HAQUE KHAN

SENIOR AI ENGINEER · BACKEND ENGINEER

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EXPERTISE

AI Systems: Fine-tuning, parameter-efficient fine-tuning (PEFT), retrieval-augmented generation (RAG), vector database, model context protocol (MCP), agentic and multimodal AI, prompt engineering, cost optimization, data safety and compliance.

Software Engineering: System architecture, backend/API development, data modeling, database integration, asynchronous processing, testing, production operations, and cloud deployment.

Engineering Leadership: Technical roadmaps, delivery planning, scoping and prioritization, code review, technical hiring, engineer onboarding and mentoring, stakeholder communication, product ownership, and cross-time-zone collaboration.

SKILLS

Languages	Python, SQL, Bash, C
GenAI	Transformers, PyTorch, LangGraph, FastMCP, llama-server, pgvector, Chroma
LLM APIs	OpenAI, Anthropic, OpenRouter, Perplexity, JinaAI, Ollama
ML & Data	scikit-learn, NLTK, OpenCV, pandas, NumPy, SciPy, Matplotlib
Backend	Django, FastAPI, Flask, Wagtail, Celery, Redis, Pytest, PostgreSQL, SQLite, Stripe
Deployment	AWS (EC2, Lightsail, S3), DigitalOcean, Render, Cloudflare (R2, Workers, D1, etc.), Caddy, Nginx
Others	REST APIs, OAuth, Webhooks, Git & GitHub, Linux, LaTeX, HTML, HTMX, Jira/ClickUp

EXPERIENCE

AI and Application Engineer

KafKai Giken Inc.

May 2023 - present

Tokyo 110-0016, Japan (Remote)

- Built and maintained the Django-PostgreSQL service and AI agents of [KAFKAI V3](#).
- Reduced LLM API costs by 50% by offloading/routing to task-specific capable models and providers.
- Built the FastMCP server, implemented action-based credit usage, and Stripe top-up for the upcoming Kafkai V4.
- Fine-tuned and maintained open-weight LLMs from HuggingFace for long-running tasks, API cost-reduction, and data safety.
- Worked on the legacy Pyramid-based codebase of Kafkai V2 and converted it to Django-PostgreSQL architecture.
- Owned production infrastructure on AWS Lightsail, managed PostgreSQL databases, Celery-Redis tasks, and deployments.
- Collaborated with UI/UX team to improve user experience of [Kafkai.com](#), [Kafkai.ai](#), and [Puchiden.app](#).
- Conducted research on product cost and margin analysis, pricing plan, and competitor studies for [Kafkai.com](#).
- Managed the affiliate and partner programs of [Puchiden.app](#).
- Conducted technical interviews and the hiring process for the Engineering team.

AI Product Engineer

ChatCopilot.app

Feb 2026 - Present

Gazipur, BD · Independent Venture

- Architected and built ChatCopilot.app, a multi-tenant Django SaaS with agentic AI based sales, support, and appointment.
- Implemented an agent runtime with tool calling, conversation state, history compaction, model routing, Bengali translation, image understanding, audio transcription, and human handover.
- Built tenant-scoped RAG for business knowledge and product/service catalogs using pgvector and multimodal embeddings.
- Designed integrations for Meta Graph API, Google Calendar, custom ERP calendars, webhooks, and Pathao courier dispatch.
- Managed the DigitalOcean production droplet, production monitoring, resource usage, and PostgreSQL database.

Research Engineer

AIMS Lab, United International University

Feb 2023 - Mar 2023

Dhaka 1212, Bangladesh

- Collaborated with the research team to transform research works into integrable applications.
- Developed the Flask backend for a multi-modal AI receptionist robot.

Machine Learning Engineer (Intern)

Shadhin Lab LLC

July 2022 - Aug 2022

Dhaka 1219, Bangladesh

- Developed NLP-based large volume data collection, labeling, and classification pipelines for an in-house product.

Undergraduate Research Assistant

Dhaka 1216, Bangladesh

Supervisor: Md. Touhidul Islam, Lecturer, Dept. of CSE, City University

Dec 2020 - Nov 2022

- Worked on machine learning research in Natural Language Processing and healthcare (leukemia, Parkinson's, CKD).
- Trained and fine-tuned machine learning and deep learning models.
- Co-authored 4 peer-reviewed publications and presented 2 accepted papers at academic conferences.

OPEN SOURCE

- **MochaChat**: Multi-character AI companion with streaming conversations.  
 - Built stateless FastAPI backend with browser-local per-character memory and streaming responses.
 - Added LLM memory compaction, multilingual support, cheap LLM routing using OpenRouter; deployed on Render.
- **Python Buddy**: A Small Language Model-based Python Coding Assistant for mobile devices. 
 - On-device inference ~10tok/s with Snapdragon680 4GB, Model Qwen-2.5-Coder-1.5-Instruct.
 - Applied PEFT, quantization. Used PyTorch, Transformers, llama.cpp.
- **Bengali AI Receptionist**: AI receptionist backend for a university admission office. 
 - Implemented face, speaker & speech recognition, RAG, text-to-speech. Used Transformers, Flask, SQLite, gTTS, OpenCV.

EDUCATION

BSc in Computer Science & Engineering

Apr 2018 - Dec 2022

City University

Dhaka 1216, Bangladesh

- **Thesis**: "Automated Detection of Acute Lymphoblastic Leukemia Subtypes from Microscopic Blood Smear Images Using Deep Neural Networks," doi: [10.48550/arXiv.2208.08992](https://doi.org/10.48550/arXiv.2208.08992)

PUBLICATIONS

- [1] **Taufiqul Haque Khan** et al., "An Intelligent Telediagnosis of Acute Lymphoblastic Leukemia using Histopathological Deep Learning," Journal of Computing Theories and Applications (JCTA), vol. 2(1), pp. 1–12, 2024. doi: [10.62411/jcta.10358](https://doi.org/10.62411/jcta.10358).
- [2] J. Hossain, M. T. Islam, and **Taufiqul Haque Khan** "Streamlining Brain Tumor Classification with Custom Transfer Learning in MRI Images," 2023 IEEE SIST, Astana, Kazakhstan, 2023, pp. 522-526, doi: [10.1109/SIST58284.2023.10223507](https://doi.org/10.1109/SIST58284.2023.10223507).
- [3] **Taufiqul Haque Khan**, M. T. Islam and F. I. Raju, "Detecting Chronic Kidney Disease (CKD) at the Initial Stage: A Novel Hybrid Feature-selection Method and Robust Data Preparation Pipeline for Different ML Techniques," 2022 5th IEEE ICCI, Cairo, Egypt, 2022, pp. 400-407, doi: [10.1109/ICCI54321.2022.9756094](https://doi.org/10.1109/ICCI54321.2022.9756094).
- [4] **Taufiqul Haque Khan** and M. T. Islam, "A Comparative Study of Sentiment Analysis Using NLP and Different ML Techniques on US Airline Twitter Data," 2021 IEEE ICECIT, Khulna, Bangladesh, 2021, pp. 1-4, doi: [10.1109/ICECIT54077.2021.9641336](https://doi.org/10.1109/ICECIT54077.2021.9641336).

VOLUNTEERING

Community Lead

March 2025 - Present

AI/ML Professional Community Bangladesh

Bangladesh

- Organize technical webinars and workshops on AI/ML applications.
- Mentor early-career practitioners in development and career pathways.

TECH EVENTS

- **Speaker**, PyCon ID 2025, Jakarta, Indonesia (14-15 December 2025)
- **Reviewer**, EuroPython 2025, Prague, Czechia (14-20 July 2025)